Lecture 12 - Oct 7

5.6 Partially Ordered Sets (Poset)

Lecture 11 - Oct 5 > Definition 2 (Binary relation)

The set X is refered as the **ground set** (the "universe")

DEFINITION (PARTIAL ORDER).

A binary relation R on a set X is a **partial order** if it is relfexive, antisymmetric, and transitive.

DEFINITION (LINEAR/TOTAL ORDER).

A binary relation R on a set X is a **linear order/total order** on X if R is reflexive, symmetric, and transitive.

$X = \mathbb{R}, R = \leq$, or the more than or equal to binary relation, is linear order since any numbers are **comparable**.

If R is a partial order on a set X, we use $\preceq$ to denote R.

THEOREM.

If $\preceq$ is a partial order on a *finite* set X, then X has a minimal element and a maximal element.

Proof.

Pick an element $a \in X$. If no $b \in X \rightarrow b \preceq a$, then we are done ($a$ is a minimal element). Otherwise, continue with b.

Since X is finite, eventually we stop at the last elemetn of the set, at which point we located the minimum element.

Similarly, we find a maximal element.

□

Question: How many minimal and maximal elements does X have if $\preceq$ is a linear order and X is finite?

Ans: unique maximal and minimal element

Proof.

Proof by contradiction.

Consider two minimal elements $a, b \in X$, then $a \preceq b$ or $b \preceq a$ must hold for both to be minimal. However, this implies $a = b$, meaning only one unique minimal element exists.

□

DEFINITION (ELEMENTS COVERING IN BINARY RELATION).

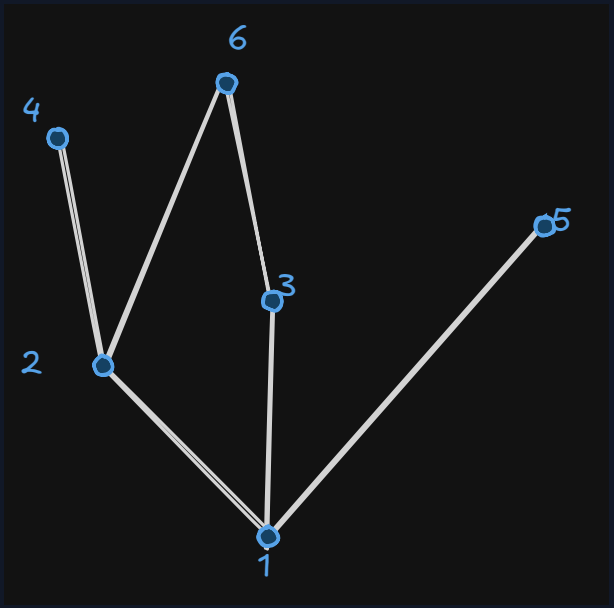
Let R be a binary relation on a set X. For $a, b \in X$, we say *a is covered by b* or *b covers a* if $\nexists c \in X$ s.t. $a \preceq c \preceq b$.

For a positive integer k, we use $[k] = \{1, 2, \ldots, k\}$.

Example

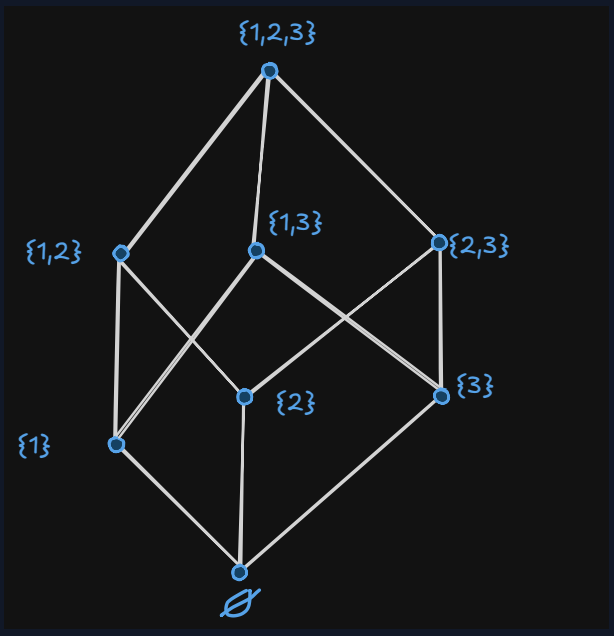
Suppose $X = [6]$, and $\preceq = \mid$ (divisibility)

The following is a **Hasse Diagram**, graphically showing the coverings of a relation



Consider another example

$X = \mathcal{P}([3]), \preceq = \subseteq$ (subsets relation)



DEFINITION (PARTIALLY ORDERED SET - POSET).

P is a pair $(X, \preceq)$, where X is a *finite* set and $\preceq$ is a partial order on X.

A poset can be visualized geometrically using a **Hasse Diagram** via cover relations.

DEFINITION (HASSE DIAGRAM).

The **Hasse Diagram** for a finite poset $P = (X, \leq)$ is obtained by taking a piont in the plane for each element in X, being careful to put the point for $v \in X$ below $y \in X$ if y *covers* v, and connecting v and y by a line segment if and only if v is covered by y.

DEFINITION (CHAIN OF A POSET).

Let $P = (X, \leq)$ be a poset.

A **chain** in the poset P is a set $C \subseteq X$, such that every pair of elements in C are comparable.

An **antichain** is a set $A \subseteq X$ such that each pair of elements in A are *incomparable*.

IF C IS A CHAIN, WHAT CAN WE SAY ABOUT THE NEW COSET $(C, \leq)$?

It now forms a linear order!

IF $P = (X, \leq)$ IS A POSET AND $\leq$ IS A LINEAR ORDER OF X, WHAT CAN WE SAY ABOUT ITS HASSE DIAGRAM?

It forms a straight line since they are all comparable

IF C IS A CHAIN AND A IS AN ANTICHAIN IN A POSET P, WHAT CAN WE KNOW ABOUT $|C \cap A|$?

$|C \cap A| \leq 1$, since the most they can share is one element

Observations: for some poset $P = (X, \leq)$:

$\{a\}$ is a chain $\forall a \in X$, since it holds the reflexive property

$\{a\}$ is an antichain $\forall a \in X$, since there are no two elements to compare

The set of all minimal elements of X form an *antichain*

The set of all maximal elements of X form an *antichain*

DEFINITION (MAXIMAL ANTICHAIN).

Given a poset $P = (X, \leq)$, a set $A \subseteq X$ is a **maximal antichain** if A is an antichain and $\nexists$ an antichain A^* such that $A \subseteq A^*$.

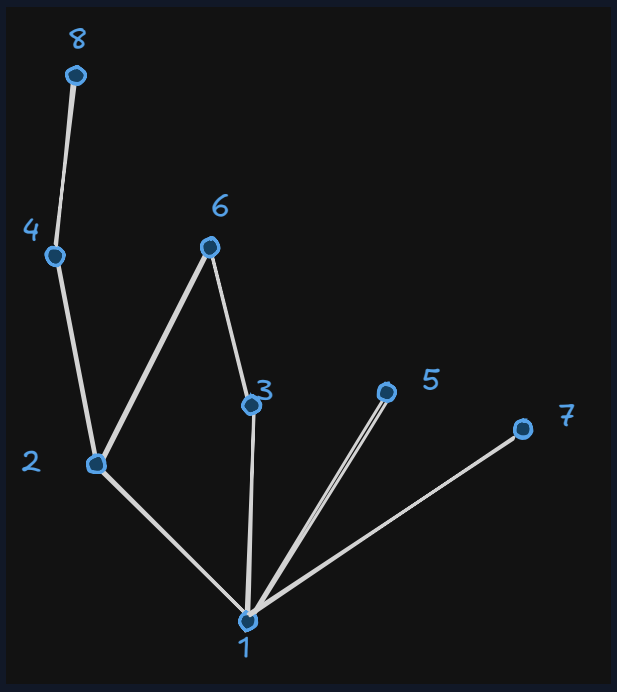
DEFINITION (MAXIMAL CHAIN).

A set is a maximal chain if c is a chiain in P and $\nexists$ a chain C^* such that $C \subseteq C^*$

Example

Consider the poset $(X, \leq)$, where $X = [8], \leq = \ |$ (divisibility)

Find the maximal chains and maximal antichains.



This forms 6 maximal chains in total:

$\{1,7\}, \{1,5\}, \{1,3,6\}, \{1,2,6\}, \{1,2,4,8\}$

Maximal antichains:

$$\{5,6,7,8\}, \{4,5,6,7\}, \{3,4,5,7\}, \{2,3,5,7\}, \{1\}, \{3,5,7,8\}$$

Questions can we partition X into chains or antichains?

Answer:

Yes, every set $\{a\}, a \in X$ forms a chain and antichain, therefore forming a partition